Config sweep — session report

srv1 / srv2, 140 cells, 20 axes

2026-08-24

1. Results

Aggregate tokens/s, measured. 475 output tokens per request with `ignore_eos`, temperature 0, one fixed prompt. **Baseline** = the configuration every prior run in this tree used: `--max-model-len 8192 --gpu-memory-utilization 0.85 --max-num-seqs 16 --enforce-eager`.

RIG	MODEL	CELLS	RAN	BASELINE	BEST	GAIN	WINNING CONFIG
srv1	Qwen2.5-Cdr-1.5B	58	41	164.3	294.7	1.79x	no-eager, len 512, seqs 256 (n=256)
srv2	Qwen2.5-Cdr-1.5B	66	51	530.1	6445.1	12.16x	no-eager, len 1024, seqs 256, kv-fp8 (n=256)
srv2	Qwen3-4B	6	6	370.0	2349.4	6.35x	no-eager, len 1024, seqs 256, kv-fp8 (n=128)
srv2	Qwen2.5-Cdr-7B	6	6	507.2	1674.3	3.30x	no-eager, len 1024, seqs 256, kv-fp8 (n=256)
srv1	Qwen2.5-Cdr-7B	4	0	-	-	-	ALL REFUSED - CUDA OOM loading weights
TOTAL		140	104				

Per-axis best, 1.5B, stage 1

Twenty axes, one factor at a time from the baseline.

AXIS	srv1	srv2	NOTE
concurrency	293.4	4006.1	the only axis that moves srv1
graphs (drop eager)	165.7	2601.7	5.02x on srv2, 0.1% on srv1
optimization-level	168.1	2354.8	rides on graphs being enabled
performance-mode	168.0	2012.7	"throughput" LOST to plain no-eager
kv-cache-dtype	167.2	592.3	inert at n=16, decisive at n=256
attention-backend	165.2	565.1	
memory (util)	167.9	564.4	
prefix-cache	167.3	561.9	
scheduler (async)	168.0	553.8	
watermark	167.8	550.9	
cascade-attn	165.4	550.2	
dtype	167.2	550.1	
batched-tokens	167.9	548.5	
stream-interval	167.0	537.4	
BASELINE	164.3	530.1	
chunked-prefill	167.8	523.5	
linear-backend	167.9	516.4	
block-size	167.9	508.9	
speculative (fixed)	201.7	2738.8	loses everywhere; -63% on srv1 at ngram-5

Decode-step cost — the mechanism

Latency divided by 475, milliseconds per decode step.

batch	srv1 1.5B	srv2 1.5B	srv2 7B (untied lm_head)	
1	22.96	27.64	17.17	
2	72.75	28.27	-	
8	74.80	28.08	15.99	<- cheaper than batch 1
32	-	-	24.25	
64	-	-	41.96	
128	95.99	28.65	81.81	

srv1 steps 3.17x between batch 1 and 2, then is flat. srv2 is flat throughout. The 7B, whose lm_head is quantized because it does not tie embeddings, has no step at all and is *cheaper* per step at batch 8 than at batch 1.

Isolated lm_head GEMM

`[151936 x hidden]`, no model loaded, same container on both rigs. Milliseconds.

	srv1 fp16	srv1 fp32	srv2 fp16
M=1	1.77	2.95	1.39
M=2	50.83	2.98	1.39
M=32	50.85	11.63	1.46

Excess on srv1 at M>=2: **49.06 ms**. Excess observed in the live vLLM run: **49.79 ms**. Agreement to 1.5%.

2.1 Knobs used

Turned — 20 axes, drawn from the engine's own 274-flag surface (`vllm serve --help=all`, 250 with printed defaults):

`max-num-seqs` · `max-model-len` · `enforce-eager` · `performance-mode` · `optimization-level` · `async-scheduling` · `dtype` · `kv-cache-dtype` · `gpu-memory-utilization` · `max-num-batched-tokens` · `block-size` · `enable-prefix-caching` · `enable-chunked-prefill` · `disable-cascade-attn` · `stream-interval` · `watermark` · `ubatch-size` · `attention-backend` · `linear-backend` · `speculative-config`

Refused by srv1 (compute-capability gates): `dtype bfloat16` · `kv-cache-dtype fp8 / fp8_e5m2 / fp8_e4m3` · `attention-backend FLASH_ATTN / FLASHINFER` · `seqs 512` · `kv-fp8 + seqs 256`

Refused by both: `linear-backend exllama / torch / machete / cutlass` · `ubatch-size 2` · `spec-method suffix` · `seqs 512`

Refused by srv2: `kv-fp8` above seqs 256

Not turned: ollama's entire surface. `OLLAMA_NUM_PARALLEL=0`, `MAX_LOADED_MODELS=0`, `KEEP_ALIVE=-1`; `OLLAMA_FLASH_ATTENTION` and `OLLAMA_KV_CACHE_TYPE` unset on both rigs.

2.2 What we learned

1. **`--enforce-eager` cost srv2 5.02x and was never required.** Justified twice in this tree as "MANDATORY on compute capability 7.5" and never once as a measurement. Worth 0.1% on the card it was claimed for; 5x at a single stream on the other (181.7 vs 36.2 tok/s at n=1), so not a batching effect.
2. **The 2026-08-19 campaign compared two misconfigured servers.** The real gap between the rigs is **21.9x**, not the 3.2x it recorded.

3. **The N=2 cliff is the unquantized lm_head.** `tie_word_embeddings: true` means lm_head is fp16 through cuBLAS, and TU116 carries no tensor cores. Microbenchmark excess 49.06 ms against 49.79 ms in vivo. Controls pass in both directions: srv1 fp32 does not cliff, srv2 fp16 does not cliff.
4. **srv1 responds to one axis of twenty.** Twenty-five cells land inside a 2.8% band; only concurrency moves it, to **~295 tok/s**, where five configurations agree within 0.4% and four context lengths agree to four significant figures.
5. **srv1's one lever is gated shut.** fp8 KV produced srv2's best cell - it doubles the sequences that fit rather than speeding a kernel up - and srv1 refuses all three fp8 variants by compute capability.
6. **fp8 KV is regime-dependent.** Inert at n=16 (558, indistinguishable from baseline), decisive at n=256 (6,445 against 6,088).
7. **Speculative decoding loses at every concurrency**, on both rigs, worse on the weaker one (-63% at ngram-5 on srv1).
8. **`--performance-mode throughput` lost** to plain no-eager, 2,012 vs 2,601.
9. **AWQ-Marlin is not sm_80-gated.** The floor is 75 and both rigs resolve `MarlinLinearKernel`. The common claim traces to vLLM #1282 from 2023.
10. **srv1 cannot hold the 7B.** CUDA OOM allocating 518 MiB with 151.88 MiB free of 5.61 GiB, in `auto_awq.py:533` during weight loading - before KV is considered. srv1 caches no untied-lm_head model it can run.
11. **The gain shrinks as the model grows** on srv2: 12.16x, 6.35x, 3.30x for 1.5B, 4B, 7B.

2.3 Still unknown / unverified

Measurement

- **The decisive mechanism test was not run.** It required an untied-lm_head model on srv1; the 7B OOMs and no other untied model is cached there. The mechanism rests on the isolated GEMM plus srv2's contrast, not on srv1 running an untied model.
- **ollama's real ceiling.** "Saturates at n=4" is a default, not a limit. Changing it needs a daemon env change and restart, and the current values are pinned by `tests/test_declared_host_state.py`.
- **3B and Qwen3-4B on srv1**, and **14B on srv2** - never swept.
- **One workload shape only**: short prompt, 475 output tokens. Nothing here speaks to prefill-heavy traffic.
- **Interactions beyond pairs.** One-factor-at-a-time plus a handful of crosses; the space is not factorially covered.
- **srv1's second step at b>8** - attributed to Marlin's narrow-M kernel (`prob_m_split <= 8`) by reading the source, not by measurement.

Instrument

- The `contract.py` constants (**#356**) - all derived under the 5x misconfiguration. START_TIMEOUT_S is directionally wrong, and RAMP_LEVELS tops out at 24 while both rigs' maxima sit at n=128-256.
- The knob surface as data (**#357**) and the resolved config entering `identity.KEY` (**#358**) - filed, not built.
- The sweep harness keeps only 25 log lines, which lost the 7B root cause until it was re-run by hand.

Errors made this session, all corrected in the record

- Concurrency arithmetic `b x max_model_len <= KV tokens` - wrong. KV is allocated per token actually used; srv1 ran 64 concurrent on a server printing `Maximum concurrency: 16.00x`.
- A shell-quoting defect produced **six false refusals** across two rigs; speculative decoding was recorded as refused when it was untested.

- Two process-management bugs (a `pgrep -f` self-match deadlock and a `pkill` self-match). Cost time, touched no data.

Landed

Evidence README and 8 result files under `records/evidence/2026-08-24-config-sweep/`. Three dated record corrections: `calibration-2026-08-19/README.md` (twice), `step0-gaps.md` gap 10, and `calibrate.py:597` in place because code is operative. Issues **#356**, **#357**, **#358** filed. Both rigs idle at 1 MiB.